Integrated Material Characterization Using SEM, FTIR, and XRF Techniques

Overview

This project involves the comprehensive characterization of material samples using multiple analytical techniques to understand their structural, chemical, and elemental properties. By combining Scanning Electron Microscopy (SEM), Fourier Transform Infrared Spectroscopy (FTIR), and X-ray Fluorescence (XRF), the study provides a multi-dimensional analysis of the material, relevant to energy, environmental, and geoscience applications.

Methodology

1. Scanning Electron Microscopy (SEM)

- Performed surface imaging under controlled conditions (~15 kV accelerating voltage)
- Examined surface morphology and microstructural features
- Conducted elemental analysis using Energy Dispersive Spectroscopy (EDS)
- Generated elemental distribution maps

2. Fourier Transform Infrared Spectroscopy (FTIR)

- Collected spectral data over a range of ~4000–500 cm^{-1}
- Identified functional groups and bonding characteristics
- Analyzed absorption peaks to determine chemical structure

3. X-ray Fluorescence (XRF)

- Performed elemental composition analysis using XRF
- Measured concentration of major and trace elements
- Identified dominant elemental components within the sample

Key Findings

SEM Analysis

- Major elements detected include Sn, O, Se, and Zn
- Surface morphology indicates a heterogeneous structure
- Elemental mapping reveals uneven distribution across the sample

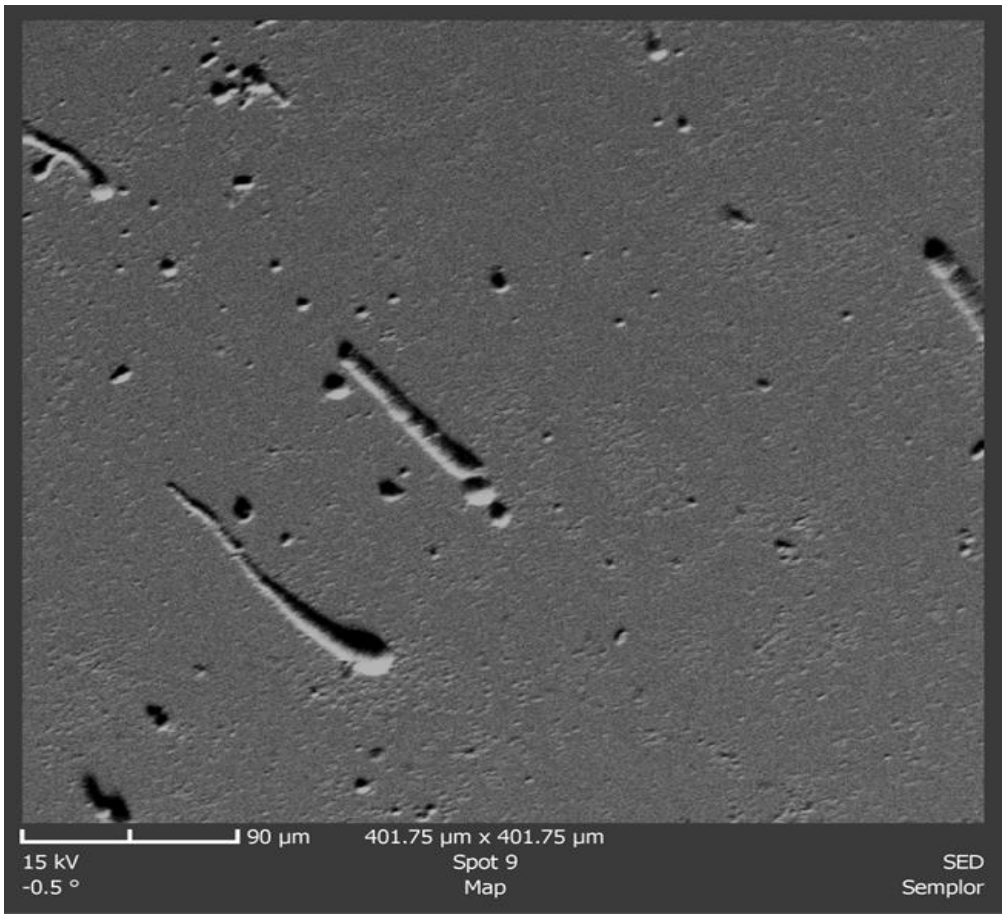

Fig 1: Surface Morphology of the sample

FTIR Analysis

- Peak at ~3277 cm^{-1} indicates O–H functional groups (possible moisture or hydroxyl bonding)
- Peaks around ~2928 cm^{-1} correspond to C–H stretching
- Absorption near ~1635 cm^{-1} suggests molecular bending vibrations
- Strong bands in 1000–1100 cm^{-1} region indicate metal–oxygen or silicate bonding
- Lower peaks (~500–800 cm^{-1}) linked to lattice vibrations

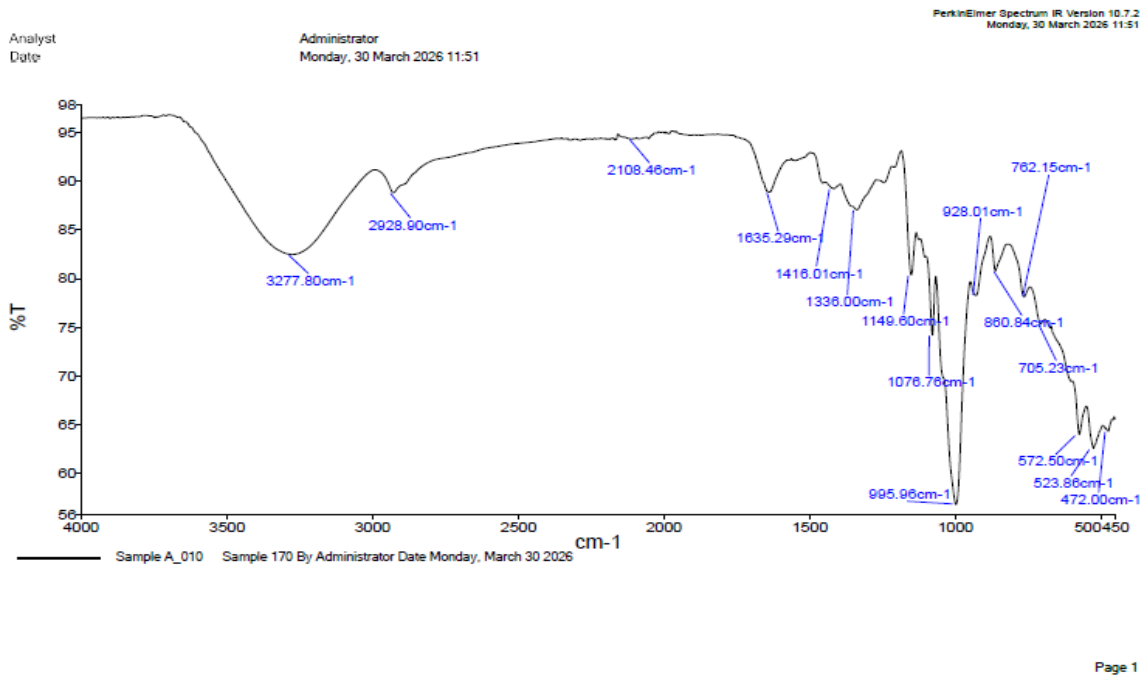

Fig 2: FTIR spectrum showing functional groups and bonding characteristics of the sample

XRF Analysis

- High concentration of sulfur (~97.6%), indicating sulfur-rich composition
- Presence of calcium, silicon, phosphorus, and iron in smaller quantities
- Trace elements such as tin (Sn) and others detected at low concentrations
- Confirms overall elemental composition and supports SEM findings

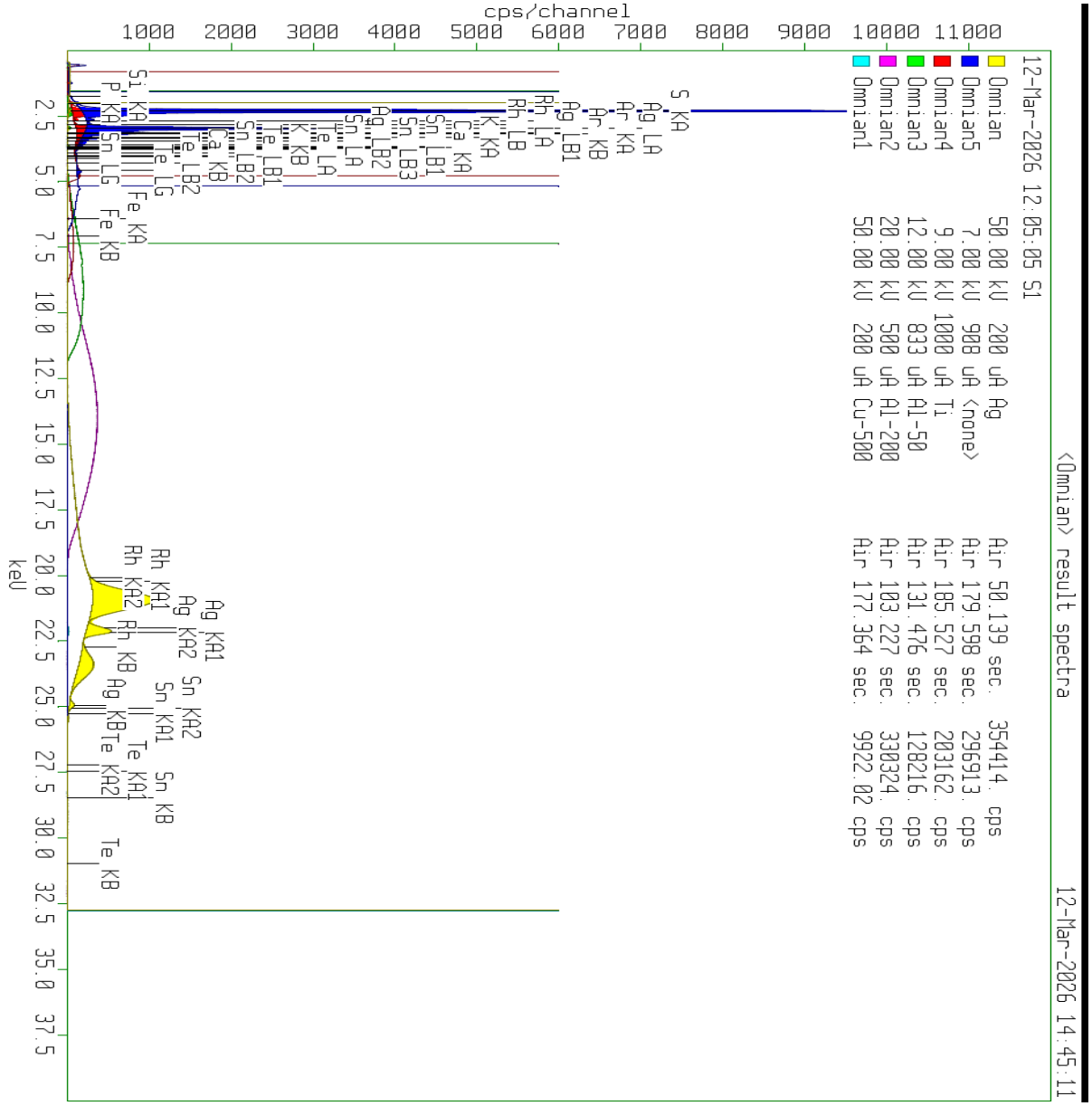

Fig 3: XRF spectrum graph representing the elemental composition of the sample

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Tools & Techniques

- Scanning Electron Microscopy (SEM)
- Energy Dispersive Spectroscopy (EDS)

- Fourier Transform Infrared Spectroscopy (FTIR)
- X-ray Fluorescence (XRF)

Relevance

This project demonstrates practical experience in advanced material characterization, integrating structural, chemical, and elemental analysis techniques. Such skills are important in geoscience, environmental studies, and energy research, particularly in:

- Mineral and material identification
- Resource evaluation
- Environmental impact assessment
- Energy material development